



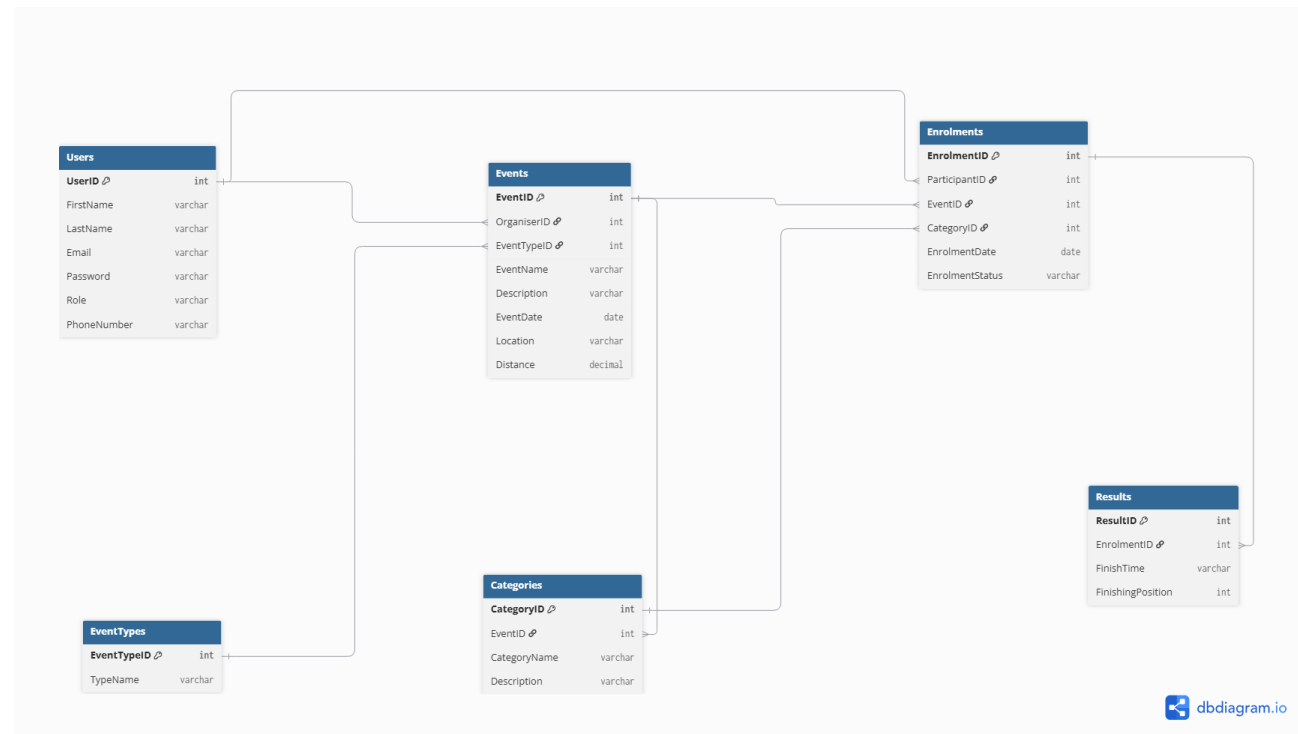
**ROSEBANK
INTERNATIONAL**

ST10489199

OMPHILE MOOPELOA

PROG PART 1

Section A



Setion B

RaceDay - Section B: API Endpoint Plan

HTTP Method	Route	Description	Role Required	Request Body	Expected Response
POST	/api/auth/register	Creates a new user account.	None	FirstName, LastName, Email, Password, PhoneNumber, Role	201 Created - User created
POST	/api/auth/login	Allows a user to log into the system.	None	Email, Password	200 OK - Login successful
GET	/api/users/profile	Allows a logged-in user to view their profile.	Any	None	200 OK - Profile details
PUT	/api/users/profile	Allows a logged-in user to update their profile.	Any	FirstName, LastName, Email, PhoneNumber	200 OK - Profile updated
GET	/api/events	Displays all available events.	None	None	200 OK - List of events
GET	/api/events/{id}	Displays information about one event.	None	None	200 OK - Event details
POST	/api/events	Creates a new event.	Organiser	EventName, Description, EventDate, Location, Distance, EventTypeID	201 Created - Event created
PUT	/api/events/{id}	Updates an existing event.	Organiser	EventName, Description, EventDate, Location, Distance, EventTypeID	200 OK - Event updated
DELETE	/api/events/{id}	Deletes an event.	Organiser	None	204 No Content - Event deleted
GET	/api/categories	Displays available event categories.	None	None	200 OK - List of categories
GET	/api/events/{id}/categories	Displays categories for a specific event.	None	None	200 OK - Categories displayed
POST	/api/events/{id}/categories	Creates a category for an event.	Organiser	CategoryName, Description	201 Created - Category created
PUT	/api/categories/{id}	Updates an event category.	Organiser	CategoryName, Description	200 OK - Category updated
DELETE	/api/categories/{id}	Deletes an event category.	Organiser	None	204 No Content - Category deleted
POST	/api/enrolments	Allows a Participant to enter an event and select a category.	Participant	EventID, CategoryID	201 Created - Enrolment created
GET	/api/enrolments/my	Allows a Participant to view their own enrolments.	Participant	None	200 OK - Participant enrolments
GET	/api/events/{id}/enrolments	Allows an Organiser to view participants enrolled in their event.	Organiser	None	200 OK - Enrolment list
POST	/api/results	Records a participant's race result.	Organiser	EnrolmentID, FinishTime, FinishingPosition	201 Created - Result recorded
GET	/api/results/my	Allows a Participant to view their own results.	Participant	None	200 OK - Participant results
GET	/api/events/{id}/results	Allows an Organiser to view results for an event.	Organiser	None	200 OK - Event results

Section C

```
-- DELETE THE OLD DATABASE IF IT ALREADY EXISTS
DROP DATABASE IF EXISTS RaceDay;
```

```
-- CREATE DATABASE
CREATE DATABASE RaceDay;
```

```
-- USE DATABASE
USE RaceDay;
```

```
-- =====
-- USERS TABLE
-- =====
```

```
CREATE TABLE Users
(
    UserID INT IDENTITY(1,1) PRIMARY KEY,
    FirstName VARCHAR(50) NOT NULL,
    LastName VARCHAR(50) NOT NULL,
    Email VARCHAR(100) NOT NULL,
    Password VARCHAR(255) NOT NULL,
    Role VARCHAR(20) NOT NULL,
    PhoneNumber VARCHAR(20)
);
```

```
-- =====
-- EVENT TYPES TABLE
-- =====
```

```
CREATE TABLE EventTypes
(
    EventTypeID INT IDENTITY(1,1) PRIMARY KEY,
    TypeName VARCHAR(50) NOT NULL
);
```

```
-- =====
-- EVENTS TABLE
-- =====
```

```

CREATE TABLE Events
(
    EventID INT IDENTITY(1,1) PRIMARY KEY,
    OrganiserID INT NOT NULL,
    EventTypeID INT NOT NULL,
    EventName VARCHAR(100) NOT NULL,
    Description VARCHAR(500),
    EventDate DATE NOT NULL,
    Location VARCHAR(100) NOT NULL,
    Distance DECIMAL(10,2) NOT NULL,

    FOREIGN KEY (OrganiserID) REFERENCES Users(UserID),
    FOREIGN KEY (EventTypeID) REFERENCES EventTypes(EventTypeID)
);

```

```

-- =====
-- CATEGORIES TABLE
-- =====

```

```

CREATE TABLE Categories
(
    CategoryID INT IDENTITY(1,1) PRIMARY KEY,
    EventID INT NOT NULL,
    CategoryName VARCHAR(50) NOT NULL,
    Description VARCHAR(200),

    FOREIGN KEY (EventID) REFERENCES Events(EventID)
);

```

```

-- =====
-- ENROLMENTS TABLE
-- =====

```

```

CREATE TABLE Enrolments
(
    EnrolmentID INT IDENTITY(1,1) PRIMARY KEY,
    ParticipantID INT NOT NULL,
    EventID INT NOT NULL,
    CategoryID INT NOT NULL,

```

```

    EnrolmentDate DATE NOT NULL,
    EnrolmentStatus VARCHAR(20) NOT NULL,

    FOREIGN KEY (ParticipantID) REFERENCES Users(UserID),
    FOREIGN KEY (EventID) REFERENCES Events(EventID),
    FOREIGN KEY (CategoryID) REFERENCES Categories(CategoryID)
);

-- =====
-- RESULTS TABLE
-- =====

CREATE TABLE Results
(
    ResultID INT IDENTITY(1,1) PRIMARY KEY,
    EnrolmentID INT NOT NULL,
    FinishTime VARCHAR(20),
    FinishingPosition INT,

    FOREIGN KEY (EnrolmentID) REFERENCES Enrolments(EnrolmentID)
);

-- =====
-- EVENT TYPES
-- =====

INSERT INTO EventTypes (TypeName)
VALUES
('Run'),
('Walk'),
('Cycle');

-- =====
-- USERS
-- =====

INSERT INTO Users
(
    FirstName,

```

```

        LastName,
        Email,
        Password,
        Role,
        PhoneNumber
    )
VALUES
(
    'Thabo',
    'Mokoena',
    'thabo@raceday.co.za',
    'Password123',
    'Organiser',
    '0712345678'
),
(
    'Lerato',
    'Dlamini',
    'lerato@raceday.co.za',
    'Password123',
    'Organiser',
    '0723456789'
),
(
    'John',
    'Smith',
    'john@gmail.com',
    'Password123',
    'Participant',
    '0734567890'
),
(
    'Nomsa',
    'Khumalo',
    'nomsa@gmail.com',
    'Password123',
    'Participant',
    '0745678901'
);

-- =====
-- EVENTS

```

```
-- =====

INSERT INTO Events
(
    OrganiserID,
    EventTypeID,
    EventName,
    Description,
    EventDate,
    Location,
    Distance
)
VALUES
(
    1,
    1,
    'Johannesburg City Run',
    'A running event through Johannesburg.',
    '2026-10-10',
    'Johannesburg',
    10.00
),
(
    2,
    2,
    'Soweto Community Walk',
    'A community walking event in Soweto.',
    '2026-10-24',
    'Soweto',
    5.00
),
(
    1,
    3,
    'Pretoria Cycle Challenge',
    'A cycling event for different experience levels.',
    '2026-11-07',
    'Pretoria',
    21.00
);

-- =====
```



```

-- CATEGORIES
-- =====

INSERT INTO Categories
(
    EventID,
    CategoryName,
    Description
)
VALUES
(
    1,
    'Under 20',
    'Participants under 20 years old.'
),
(
    1,
    'Senior',
    'Adult participants.'
),
(
    2,
    '5km Open',
    'Open category for the 5km walk.'
),
(
    2,
    'Family',
    'Category for family participants.'
),
(
    3,
    '21km Open',
    'Open category for the 21km cycle.'
),
(
    3,
    'Senior',
    'Senior cycling category.'
);

-- =====

```

```
-- ENROLMENTS
-- =====

INSERT INTO Enrolments
(
    ParticipantID,
    EventID,
    CategoryID,
    EnrolmentDate,
    EnrolmentStatus
)
VALUES
(
    3,
    1,
    2,
    '2026-09-01',
    'Confirmed'
),
(
    4,
    1,
    2,
    '2026-09-02',
    'Confirmed'
),
(
    3,
    2,
    3,
    '2026-09-03',
    'Pending'
),
(
    4,
    3,
    5,
    '2026-09-03',
    'Confirmed'
);

-- =====
```

```

-- RESULTS
-- =====

INSERT INTO Results
(
    EnrolmentID,
    FinishTime,
    FinishingPosition
)
VALUES
(
    1,
    '00:52:30',
    47
),
(
    2,
    '00:55:15',
    62
);

SELECT
    Users.FirstName,
    Users.LastName,
    Events.EventName,
    Categories.CategoryName
FROM Users
INNER JOIN Events
    ON Users.UserID = Events.OrganiserID
INNER JOIN Categories
    ON Events.EventID = Categories.EventID;

SELECT
    Users.FirstName,
    Users.LastName,
    Events.EventName,
    Events.EventDate
FROM Users
INNER JOIN Events
    ON Users.UserID = Events.OrganiserID;

SELECT
    Events.EventName,
    EventTypes.TypeName,
    Events.EventDate,

```

```

        Events.Location
FROM Events
INNER JOIN EventTypes
    ON Events.EventTypeID = EventTypes.EventTypeID;

SELECT
    Users.FirstName,
    Users.LastName,
    Events.EventName,
    Categories.CategoryName
FROM Users
INNER JOIN Events
    ON Users.UserID = Events.OrganiserID
INNER JOIN Categories
    ON Events.EventID = Categories.EventID;

SELECT
    Users.FirstName,
    Users.LastName,
    Enrolments.EventID,
    Enrolments.EnrolmentDate,
    Enrolments.EnrolmentStatus
FROM Users
INNER JOIN Enrolments
    ON Users.UserID = Enrolments.ParticipantID;

SELECT
    Enrolments.EnrolmentID,
    Events.EventName,
    Enrolments.EnrolmentDate,
    Enrolments.EnrolmentStatus
FROM Enrolments
INNER JOIN Events
    ON Enrolments.EventID = Events.EventID;

SELECT
    Enrolments.EnrolmentID,
    Categories.CategoryName,
    Enrolments.EnrolmentStatus
FROM Enrolments
INNER JOIN Categories
    ON Enrolments.CategoryID = Categories.CategoryID;

SELECT

```

```

    Users.FirstName,
    Users.LastName,
    Events.EventName,
    Enrolments.EnrolmentDate,
    Enrolments.EnrolmentStatus
FROM Users
INNER JOIN Enrolments
    ON Users.UserID = Enrolments.ParticipantID
INNER JOIN Events
    ON Enrolments.EventID = Events.EventID;

SELECT
    Users.FirstName,
    Users.LastName,
    Events.EventName,
    Categories.CategoryName,
    Enrolments.EnrolmentDate,
    Enrolments.EnrolmentStatus
FROM Users
INNER JOIN Enrolments
    ON Users.UserID = Enrolments.ParticipantID
INNER JOIN Events
    ON Enrolments.EventID = Events.EventID
INNER JOIN Categories
    ON Enrolments.CategoryID = Categories.CategoryID;

SELECT
    Enrolments.EnrolmentID,
    Enrolments.ParticipantID,
    Results.FinishTime,
    Results.FinishingPosition
FROM Enrolments
INNER JOIN Results
    ON Enrolments.EnrolmentID = Results.EnrolmentID;

SELECT
    Users.FirstName,
    Users.LastName,
    Results.FinishTime,
    Results.FinishingPosition
FROM Users
INNER JOIN Enrolments
    ON Users.UserID = Enrolments.ParticipantID
INNER JOIN Results

```

```
    ON Enrolments.EnrolmentID = Results.EnrolmentID;

-- =====
-- DISPLAY ALL TABLES
-- =====

SELECT * FROM Users;

SELECT * FROM EventTypes;

SELECT * FROM Events;

SELECT * FROM Categories;

SELECT * FROM Enrolments;

SELECT * FROM Results;
```